CS4186: Vision & Image

Image Resampling , Interpolation

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Image scaling

This image is too big to fit on the screen. How can we generate a half-sized version?

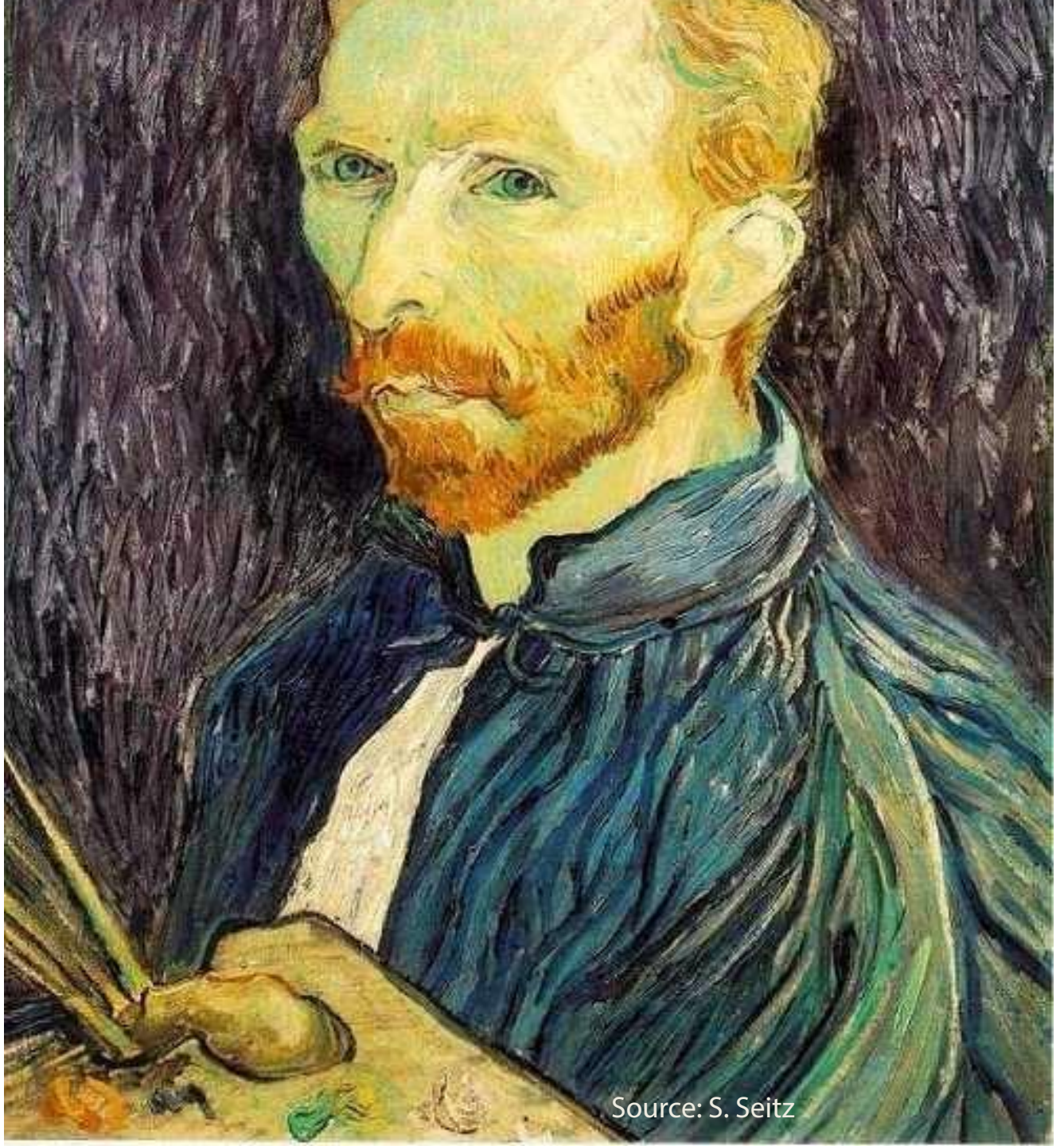
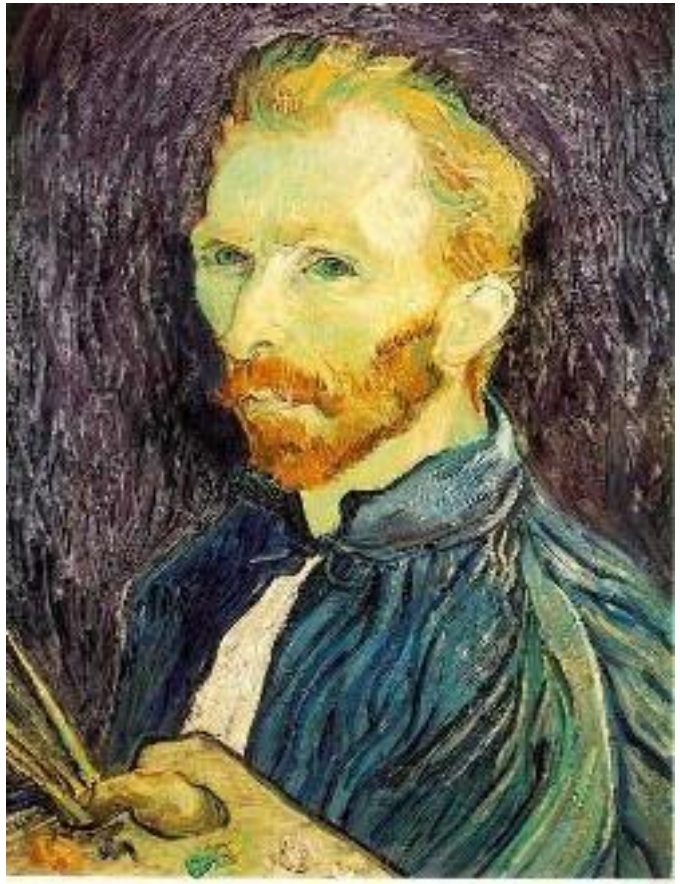


Image sub-sampling



Throw away every other row and column to create a 1/2 size image - called *image sub-sampling*



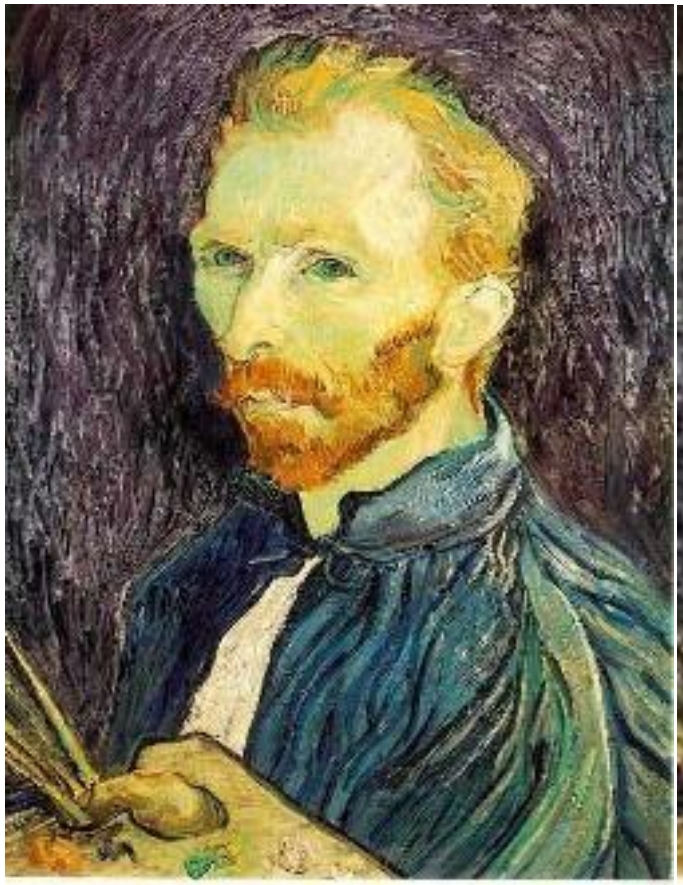
1/4



1/8

Source: S. Seitz

Image sub-sampling



1/2



1/4 (2x zoom)



1/8 (4x zoom)

Why does this look so crufty?

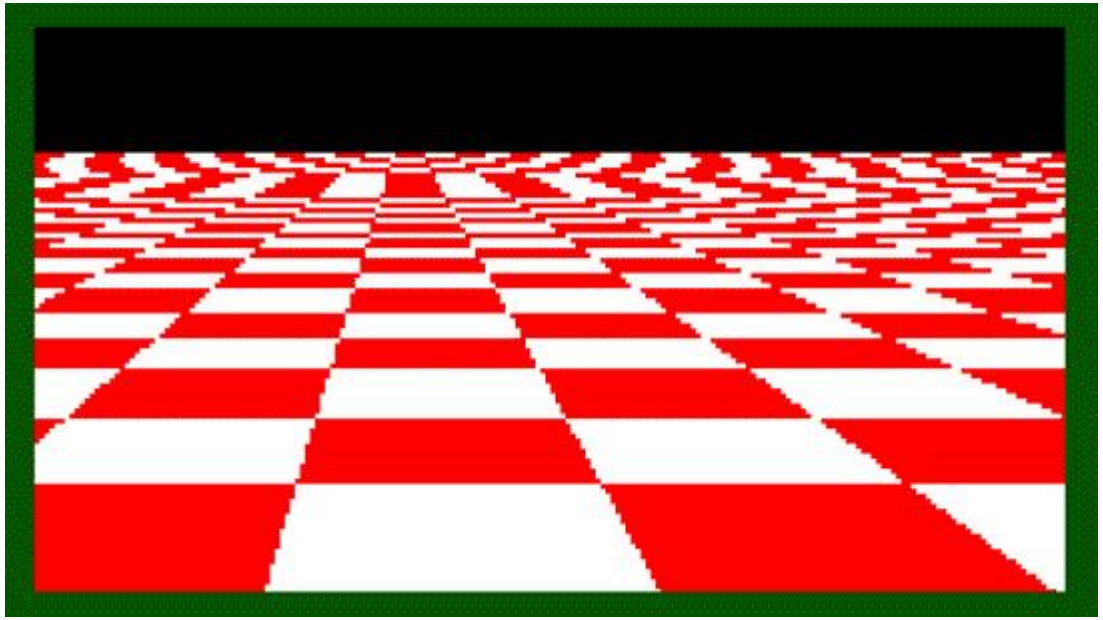
Source: S. Seitz

Image sub-sampling – another example



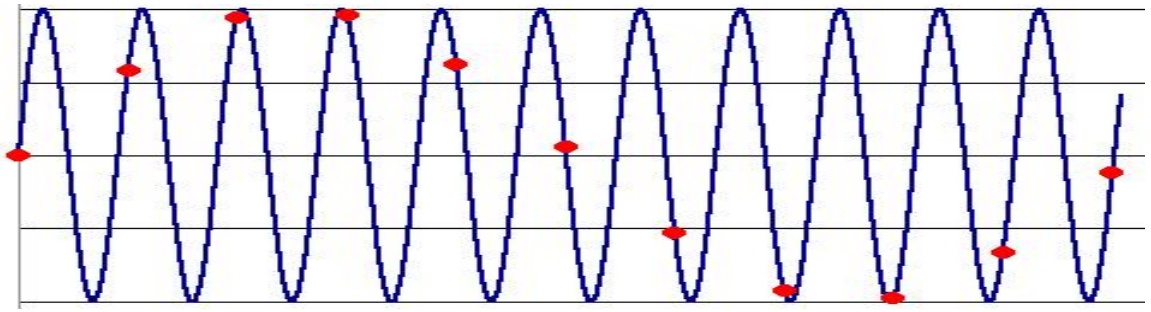
Source: F. Durand

Even worse for synthetic images



Source: L. Zhang

Aliasing



- Occurs when your sampling rate is not high enough to capture the amount of detail in your image
- Can give you the wrong signal/image—an *alias*

- To do sampling right, need to understand the structure of your signal/image
- Enter Monsieur Fourier…
  - "But what is the Fourier Transform? A visual introduction."

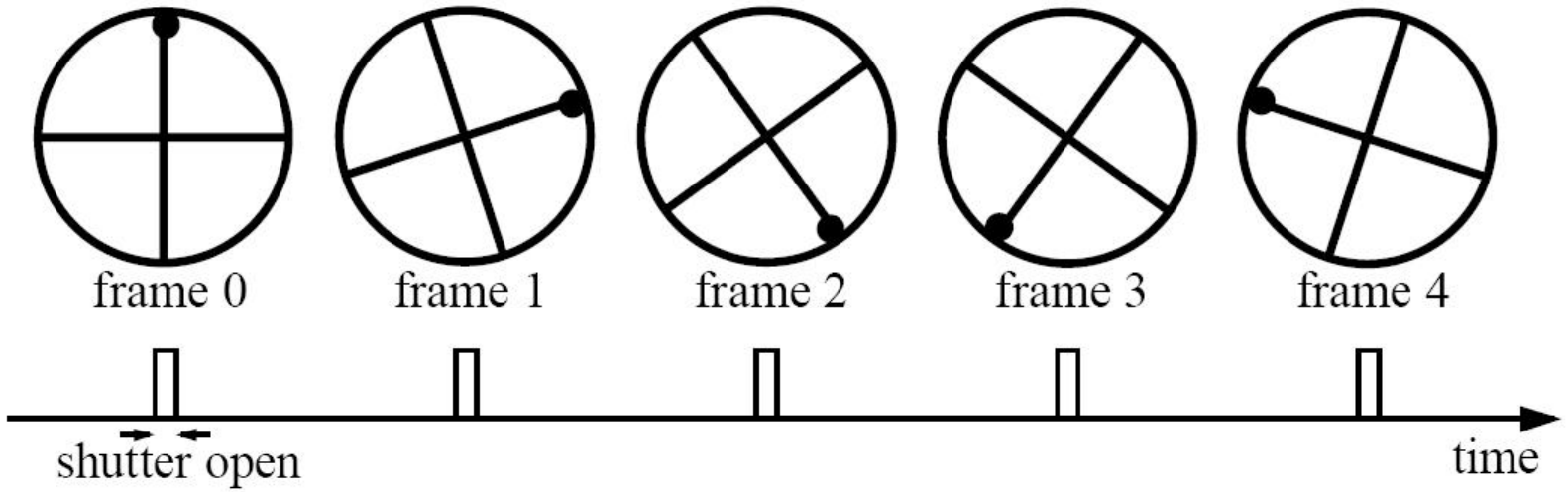
  To avoid aliasing:
  - sampling rate ≥ 2 * max frequency in the image
    - said another way: ≥ two samples per cycle
  - This minimum sampling rate is called the **Nyquist rate**

Source: L. Zhang

Wagon-wheel effect

Imagine a spoked wheel moving to the right (rotating clockwise). Mark wheel with dot so we can see what's happening.

If camera shutter is only open for a fraction of a frame time (frame time = 1/30 sec. for video, 1/24 sec. for film):



Without dot, wheel appears to be rotating slowly backwards! (counterclockwise)

(See http://www.michaelbach.de/ot/mot-wagonWheel/index.html)

Source: L. Zhang